

MonoAI MONAI Security Audit Report

Defensive cybersecurity and integrity review for MONAI-based medical AI projects.

1. Scope

Field	Value
Project root	C:\Users\kozakka\monai_test\audit_cyber_monoai\monoai_demo_projects
Dataset root	C:\Users\kozakka\monai_test\audit_cyber_monoai\monoai_demo_projects\data
Model path	C:\Users\kozakka\monai_test\audit_cyber_monoai\monoai_demo_projects\models
Platform	Windows-10-10.0.26200-SP0

2. Executive Summary

Metric	Count
Dependencies scanned	7
Model files scanned	0
Dataset files scanned	0
MONAI transforms detected	55
Issues total	35
CRITICAL	0
ERROR	2
WARNING	0
INFO	33

3. Issues

Level	Scanner	Category	Message
ERROR	model_file_scanner	missing_model_file	Model file does not exist.
ERROR	dataset_metadata_scanner	missing_dataset_root	Dataset root does not exist.
INFO	monai_transform_scanner	transform_review	LoadImaged: Verify file paths are controlled and not user-supplied without validation.
INFO	monai_transform_scanner	transform_review	Orientationd: Orientation changes coordinate convention. Confirm image and label tra
INFO	monai_transform_scanner	transform_review	Spacingd: Spacing changes voxel geometry. Confirm spacing is logged and consiste
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4. Dependency Scanner

Package	Version
monai	1.5.2
torch	2.10.0
numpy	2.2.6
nibabel	5.4.2
scipy	1.15.3
pydicom	3.0.1
matplotlib	3.10.8

5. Model File Scanner

No model files scanned.

6. Dataset Metadata Scanner

No dataset metadata scanned.

7. MONAI Transform Scanner

File	Line	Transform	Keys	Notes
C:\Users\kozakka\monai_test\audit_cyber6\monai...		LoadImaged	['image', 'label']	Verify file paths are controlled and not user-supplied without va
C:\Users\kozakka\monai_test\audit_cyber7\monai...		EnsureChannelFirstd	['image', 'label']	
C:\Users\kozakka\monai_test\audit_cyber8\monai...		Orientationd	['image', 'label']	Orientation changes coordinate convention. Confirm image and
C:\Users\kozakka\monai_test\audit_cyber9\monai...		Spacingd	['image', 'label']	Spacing changes voxel geometry. Confirm spacing is logged a
C:\Users\kozakka\monai_test\audit_cyber10\monai...		ScaleIntensityd	['image']	
C:\Users\kozakka\monai_test\audit_cyber6\monai...		LoadImaged	['image', 'label']	Verify file paths are controlled and not user-supplied without va
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C:\Users\kozakka\monai_test\audit_cyber9\monai...		Spacingd	['image', 'label']	Spacing changes voxel geometry. Confirm spacing is logged a
C:\Users\kozakka\monai_test\audit_cyber10\monai...		ScaleIntensityd	['image']	
C:\Users\kozakka\monai_test\audit_cyber6\monai...		LoadImaged	['image', 'label']	Verify file paths are controlled and not user-supplied without va
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C:\Users\kozakka\monai_test\audit_cyber10_monoai\...		ScaleIntensityd	['image']	

8. Defensive Recommendations

- Keep model artifacts and datasets checksum-verified.
- Avoid loading untrusted pickle-based PyTorch checkpoints.
- Use reproducible dependency pinning for clinical and regulated workflows.
- Audit NIfTI affine, spacing, label class values, and image-label shape consistency.
- Review MONAI random transforms and set deterministic seeds for validation and audits.
- Store PDF, Markdown, JSON, and CSV audit reports with experiment artifacts.